Conduction, convection, radiation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify mechanisms of heat transfer and support each classification.

SUCCESS CRITERIA - I can...

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use evidence from the situation.
3. Recognize that more than one mechanism can act.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “conduction” and “convection” to explain your model or result. Sentence starters are optional.

conduction: energy transfer through microscopic interactions in matter

convection: energy transfer through bulk motion of a fluid

radiation: energy transfer by electromagnetic waves

01 Bring back an earlier idea

A device transfers 15 J in 3 s, then performs the same transfer in 6 s.
Find both average powers and explain the difference.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

Worked reasoning: A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

Your turn: Draw one distinct, labeled model for each mechanism.
Show the direction of energy transfer.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand.
Give a reason for each.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

04 Build the solution

Draw arrows showing warm fluid rising and cooler fluid sinking in a simple convection loop.
Explain why this is not the same as conduction along a solid spoon.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Sentence starters are optional. You may plan in any language.

05 Use the idea

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A rolling object loses 24 J gravitational energy but gains only 16 J translational kinetic energy.
Name two possible destinations of the difference.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Explain how energy can travel from the Sun to Earth through space.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design insulation for a hot cup using one change for each transfer mechanism.
Explain each change.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3